Pairs of $f(x)$ & $f'(x)$

① $$\int f'(x)[f(x)]^n \, dx = \frac{[f(x)]^{n+1}}{n+1} + c \quad , \boxed{n \neq -1}$$

② $$\int \frac{f'(x)}{f(x)} \, dx = \ln|f(x)| + c$$

③ $$\int f'(x) e^{f(x)} \, dx = e^{f(x)} + c$$

"x"

① $$\int (ax+b)^n \, dx = \frac{(ax+b)^{n+1}}{a(n+1)} + c \quad , \boxed{\begin{array}{l} n \neq -1 \\ a \neq 0 \end{array}}$$

② $$\int \frac{1}{ax+b} \, dx = \frac{1}{a} \ln|ax+b| + c \quad , \boxed{a \neq 0}$$

If $a = 1$,

$$\Rightarrow \int \frac{1}{x+b} \, dx = \ln|x+b| + c$$

" x^2 "

① $$\int \frac{1}{x^2 + a^2}\,dx = \frac{1}{a}\tan^{-1}\frac{x}{a} + c$$

"e"

If $a = 1$, $b = 0$

$\Rightarrow \int e^x\,dx = e^x$

① $$\int e^{ax+b}\,dx = \frac{1}{a}e^{ax+b} + c$$

$\boxed{a \neq 0}$

---

② $$\int f'(x)\,e^{f(x)}\,dx = e^{f(x)} + c$$

Sin x

① $$\int \sin(ax+b)\,dx = -\frac{1}{a}\cos(ax+b) + c$$

If $a = 1$,

$\int \sin(x+b)\,dx = -\cos(x+b) + c$

$\boxed{a \neq 0}$

$\cos x$

① $$\int \cos(ax+b)\,dx = \frac{1}{a}\sin(ax+b) + C$$

If $a = 1$,

$$\int \cos(x+b)\,dx = \sin(x+b) + C$$

$\boxed{a \neq 0}$

$\sec^2 x$

① $$\int \sec^2(ax+b)\,dx = \frac{1}{a}\tan(ax+b) + C$$

If $a = 1$,

$$\int \sec^2(x+b)\,dx = \tan(x+b) + C$$

$\boxed{a \neq 0}$

Pairs of $f(x)$ & $f'(x)$

① $\int f'(x) \sin f(x) \, dx = -\cos f(x) + c$

---

② $\int f'(x) \cos f(x) \, dx = \sin f(x) + c$

---

③ $\int f'(x) \sec^2 f(x) \, dx = \tan f(x) + c$

Common Tricks

$\int \sin^2 x$

$$\int \sin^2 x \, dx$$

$$= \int \frac{1}{2}(1-\cos 2x)\, dx$$

$$= \frac{1}{2}\int (1-\cos 2x)\, dx$$

$$= \frac{1}{2}\left(x - \frac{1}{2}\sin 2x\right) + c$$

$$= \frac{1}{2}x - \frac{1}{4}\sin 2x + c$$

We know that

$\cos 2x = 1 - 2\sin^2 x$

$\Rightarrow \sin^2 x = \frac{1}{2}(1-\cos 2x)$

$\int \cos^2 x$

$$\int \cos^2 x \, dx$$

$$= \int \frac{1}{2}(1+\cos 2x)\, dx$$

$$= \frac{1}{2}\int (1+\cos 2x) + c$$

$$= \frac{1}{2}\left(x + \frac{1}{2}\sin 2x\right) + c$$

$$= \frac{1}{2}x + \frac{1}{4}\sin 2x + c$$

We know that

$\cos 2x = 2\cos^2 x - 1$

$\Rightarrow \cos^2 x = \frac{1}{2}(1+\cos 2x)$

$\int \sin^3 x$

$$\int \sin^3 x \, dx$$

$$= \int (\sin^2 x)(\sin x)\, dx$$

$$= \int (1-\cos^2 x)(\sin x)\, dx$$

We know that
$\sin^2 x + \cos^2 x = 1$
$\Rightarrow \sin^2 x = 1 - \cos^2 x$

$$= \int \sin x + (\underbrace{-\sin x}_{f'(x)})(\underbrace{\cos x}_{f(x)})^2 \, dx$$

$$= -\cos x + \frac{(\cos x)^3}{3} + c$$

$$= \frac{1}{3}\cos^3 x - \cos x + c \quad \#$$

OR

let $u = \cos x$
$du = -\sin x \, dx$

$$= -\int (1-u^2)\, du$$

$$= -\left[u - \frac{u^3}{3}\right]$$

$$= -u + \frac{u^3}{3}$$

$$= -\cos x + \frac{\cos^3 x}{3}$$

$$= \frac{1}{3}\cos^3 x - \cos x + C \quad \#$$

$\int \cos^3 x$

$$\int \cos^3 x \, dx$$

$$= \int (\cos^2 x)(\cos x)\, dx$$

$$= \int (1-\sin^2 x)(\cos x)\, dx$$

We know that
$\sin^2 x + \cos^2 x = 1$
$\Rightarrow \cos^2 x = 1 - \sin^2 x$

$$= \int \cos x - (\underbrace{\cos x}_{f'(x)})(\underbrace{\sin x}_{f(x)})^2 \, dx$$

$$= \sin x - \frac{(\sin x)^3}{3}$$

$$= -\frac{1}{3}\sin^3 x + \sin x + c \quad \#$$

OR

Let $u = \sin x$
$du = \cos x \, dx$

$$= \int (1-u^2)\, du$$

$$= u - \frac{u^3}{3} + c$$

$$= \sin x - \frac{\sin^3 x}{3} + c$$

$$= -\frac{1}{3}\sin^3 x + \sin x + c$$

$\int \sin^4 x$

$$\int \sin^4 x \, dx$$

$$= \int [\sin^2 x]^2 \, dx$$

$$= \int \left[\tfrac{1}{2}(1-\cos 2x)\right]^2 dx$$

$$= \tfrac{1}{4} \int (1-\cos 2x)^2 \, dx$$

$$= \tfrac{1}{4} \int [1 - 2\cos 2x + \cos^2 2x] \, dx$$

$$= \tfrac{1}{4} \int \left[1 - 2\cos 2x + \tfrac{1}{2}(1+\cos 4x)\right] dx$$

$$= \tfrac{1}{4} \int \left[\tfrac{3}{2} - 2\cos 2x + \tfrac{1}{2}\cos 4x\right] dx$$

$$= \tfrac{1}{4}\left(\tfrac{3}{2}x - \tfrac{2}{2}\sin 2x - \tfrac{1}{8}\sin 4x\right)$$

$$= \tfrac{3}{8}x - \tfrac{1}{4}\sin 2x + \tfrac{1}{32}\sin 4x + C$$ #

$\int \cos^4 x$

$$\int \cos^4 x \, dx$$

$$= \int [\cos^2 x]^2 \, dx$$

$$= \int \left[\tfrac{1}{2}(1+\cos 2x)\right]^2 dx$$

$$= \tfrac{1}{4} \int (1+\cos 2x)^2 \, dx$$

$$= \tfrac{1}{4} \int [1 + 2\cos 2x + \cos^2 2x]$$

$$\tfrac{1}{4} \int \left[1 + 2\cos 2x + \tfrac{1}{2}(1+\cos 4x)\right] dx$$

$$= \tfrac{1}{4} \int \left[1 + 2\cos 2x + \tfrac{1}{2} + \tfrac{1}{2}\cos 4x\right] dx$$

$$= \tfrac{1}{4} \int \left[\tfrac{3}{2} + 2\cos 2x + \tfrac{1}{2}\cos 4x\right] dx$$

$$= \tfrac{1}{4}\left[\tfrac{3}{2}x + \sin 2x + \tfrac{1}{8}\sin 4x\right]$$

$$= \tfrac{3}{8}x + \tfrac{1}{4}\sin 2x + \tfrac{1}{32}\sin 4x + C$$ #

$$\boxed{\int \tan^2 x}$$

$$\int \tan^2 x \, dx$$

We know that

$$1 + \tan^2 x = \sec^2 x$$

$$\Rightarrow \tan^2 x = \sec^2 x - 1$$

$$= \int (\sec^2 x - 1) \, dx$$

$$= \tan x - x + c$$

$$\boxed{\int \tan x}$$

$$\int \tan x \, dx$$

$$= \int \frac{\sin x}{\cos x} \, dx$$

$$= -\int \frac{-\sin x}{\cos x}$$

$$\int \frac{f'(x)}{f(x)} = \ln|f(x)| + c$$

$$= -\ln|\cos x| + c$$

$$\boxed{\int \cot x}$$

$$\int \cot x \, dx$$

$$= \int \frac{1}{\tan x} \, dx$$

$$= \int \frac{\cos x}{\sin x} \, dx$$

$$= \ln|\sin x| + c$$

$$\boxed{\int \frac{f'(x)}{f(x)} = \ln|f(x)| + c}$$

Rational Fractions

Improper

$\Rightarrow \dfrac{f(x)}{g(x)}$, Deg $f(x) \geq$ Deg $g(x)$

[Do long division first]

Example 1:

$$\int \frac{x^2+1}{2-3x}\,dx$$

$$\begin{array}{r}
-\frac{1}{3}x - \frac{2}{9} \\
-3x+2 \,\big)\, \overline{x^2 + 0x + 1} \\
-x^2 - \frac{2}{3}x \\
\hline
\frac{2}{3}x + 1 \\
\frac{2}{3}x - \frac{4}{9} \\
\hline
\frac{13}{9}
\end{array}$$

$$\therefore \frac{x^2+1}{2-3x} = \left(-\frac{1}{3}x - \frac{2}{9}\right) + \frac{13/9}{-3x+2}$$

$$\int \frac{x^2+1}{2-3x} = \int \left[-\frac{1}{3}x - \frac{2}{9} + \frac{13/9}{-3x+2}\right] dx$$

$$= -\frac{1}{6}x^2 - \frac{2}{9}x + \frac{13}{9}\left(-\frac{1}{3}\right)\ln|-3x+2| + c$$

$\boxed{\text{Proper}} \Rightarrow \frac{f(x)}{g(x)}$, Deg $f(x) <$ Deg $g(x)$

[Do partial fraction]

Example 1:

$$\int \frac{2x}{(x-1)^2(x+2)}\,dx$$

Let $\frac{2x}{(x-1)^2(x+2)} = \frac{A}{(x-1)} + \frac{B}{(x-1)^2} + \frac{C}{(x+2)}$

$$2x = A(x-1)(x+2) + B(x+2) + C(x-1)^2$$

Let $x = 1 \rightarrow 2 = B(1+2)$

$$\boxed{B = \frac{2}{3}}$$

Let $x = -2 \rightarrow -4 = C(-2-1)^2$

$$\boxed{C = -\frac{4}{9}}$$

Let $x = 0 \longrightarrow 0 = A(-1)(2) + B(2) + C(-1)^2$

$$0 = -2A + \left(\frac{2}{3}\right)(2) + \left(-\frac{4}{9}\right)(1)$$

$$\boxed{A = \frac{4}{9}}$$

$$\therefore \frac{2x}{(x-1)^2(x+2)} = \frac{4/9}{(x-1)} + \frac{2/3}{(x-1)^2} - \frac{4/9}{(x+2)}$$

$$\therefore \int \frac{2x}{(x-1)^2(x+2)} = \int \left[\frac{4/9}{(x-1)} + \frac{2/3}{(x-1)^2} - \frac{4/9}{(x+2)}\right] dx$$

$$= \int \left[\frac{4/9}{(x-1)} + \frac{2}{3}(x-1)^{-2} - \frac{4/9}{(x+2)}\right] dx$$

cont. below

↓

$$= \frac{4}{9}\ln|x-1| + \frac{2}{3}\frac{(x-1)^{-1}}{-1(1)} - \frac{4}{9}\ln|x+2| + c$$

$$= \frac{4}{9}\ln|x-1| - \frac{2}{3(x-1)} - \frac{4}{9}\ln|x+2| + c$$

$$= \frac{4}{9}\left[\ln|x-1| - \ln|x+2|\right] - \frac{2}{3(x-1)} + c \quad \#$$

Irreducible denominator

$\Rightarrow b^2 - 4ac < 0$

[Complete the squares]

Example 1:

$$\int \frac{1}{x^2+4x+7}\,dx$$

$x^2 + 4x + 7$

$= x^2 + 4x + \left(\frac{4}{2(1)}\right)^2 - \left(\frac{4}{2(1)}\right)^2 + 7$

$= x^2 + 4x + (2)^2 - (2)^2 + 7$

$= (x+2)^2 + 3$

$$= \int \frac{1}{(x+2)^2 + 3}\,dx$$

Let $u = x+2$

$\frac{du}{dx} = 1$

$du = dx$

$$= \int \frac{1}{u^2 + (\sqrt{3})^2}\,du$$

$$= \frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{u}{\sqrt{3}}\right) + c$$

$$= \frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{x+2}{\sqrt{3}}\right) + c \quad \#$$

Integration by parts

Stop 1 : When u becomes 0

Example 1

$$\int 2x\, e^{x}\, dx$$

$$= 2xe^{x} - \int 2e^{x}$$

$$= 2xe^{x} - 2e^{x}$$

$u \rightarrow \frac{du}{dx}$ $\quad$ $\frac{dv}{dx} \rightarrow v$

	$\boxed{u}$ D	$\boxed{\frac{dv}{dx}}$ I
+	$2x$	e^{x}
−	2	e^{x}
+	0	e^{x}

Example 2

$$\int x^{2} e^{3x}\, dx$$

$$= \frac{1}{3}x^{2}e^{3x} - \int \frac{2}{3}xe^{3x}$$

$$= \frac{1}{3}x^{2}e^{3x} - \left[\frac{2}{9}xe^{3x} - \int \frac{2}{9}e^{3x}\right]$$

$$= \frac{1}{3}x^{2}e^{3x} - \frac{2}{9}xe^{3x} + \int \frac{2}{9}e^{3x}$$

$$= \frac{1}{3}x^{2}e^{3x} - \frac{2}{9}xe^{3x} + \frac{2}{27}e^{3x} + c \ \#$$

$u \rightarrow \frac{du}{dx}$ $\quad$ $\frac{dv}{dx} \rightarrow v$

	$\boxed{u}$ D	$\boxed{\frac{dv}{dx}}$ I
+	x^{2}	e^{3x}
−	$2x$	$\frac{1}{3}e^{3x}$
+	2	$\frac{1}{9}e^{3x}$
−	0	$\frac{1}{27}e^{3x}$

Stop 2 : When row can be integrated

Example 1 :

$$\int x^2 \ln x \, dx$$

$$= \frac{1}{3}x^3 \ln x - \int \left(\frac{1}{x}\right)\left(\frac{x^3}{3}\right) dx$$

$$= \frac{1}{3}x^3 \ln x - \int \frac{x^2}{3} \, dx$$

$$= \frac{1}{3}x^3 \ln x - \frac{x^3}{9} + c \quad dx$$

	$\boxed{u}$ D	$\boxed{\frac{dv}{dx}}$ I	
	$+ \ln x$	x^2	
$u \rightarrow \frac{du}{dx}$	$- \frac{1}{x}$	$\frac{1}{3}x^3$	$\frac{dv}{dx} \rightarrow v$
$u \rightarrow \frac{du}{dx}$	$+ \left(-\frac{1}{2x}\right)$	$\frac{1}{12}x^4$	$\frac{dv}{dx} \rightarrow v$

Step 3: When row same as original

Example 1:

$$\int e^{3x} \sin x \, dx$$

	$\boxed{u}$ D	$\boxed{\frac{dv}{dx}}$ I
$+$	e^{3x}	$\sin x$
$-$	$3e^{3x}$	$-\cos x$
$+$	$9e^{3x}$	$-\sin x$
$-$	$27e^{3x}$	$\cos x$

$u \rightarrow \frac{du}{dx}$

$\frac{dv}{dx} \rightarrow v$

$$\int e^{3x} \sin x = -e^{3x} \cos x - \int -3e^{3x} \cos x \, dx$$

$$\int e^{3x} \sin x = -e^{3x} \cos x + 3e^{3x} \sin x + \int -9e^{3x} \sin x \, dx$$

$$\int e^{3x} \sin x = -e^{3x} \cos x + 3e^{3x} \sin x - 9\int e^{3x} \sin x \, dx$$

$$10\int e^{3x} \sin x = -e^{3x} \cos x + 3e^{3x} \sin x + c$$

$$\int e^{3x} \sin x = -\frac{1}{10} e^{3x} \cos x + \frac{3}{10} e^{3x} \sin x + \frac{c}{10}$$

$$= -\frac{1}{10} e^{3x} \cos x + \frac{3}{10} e^{3x} \sin x + A, \text{ where } A = \frac{c}{10}$$